

Expertise and Resources — Not Anonymized

Mapping Reionization-Era Ionized Bubbles by Recovering Lyman- α Information from Low-Resolution JWST and Roman Spectroscopy

1. Team Members, Affiliations, and Roles

Name	Affiliation	Role
Prof. Bahram Mobasher	University of California, Riverside	PI
Aryana Haghjoo	University of California, Riverside	FI (PhD student)
Dr. Shoubaneh Hemmati	IPAC, California Institute of Technology	Collaborator (Mentor 1)

2. Team Expertise and Responsibilities

Aryana Haghjoo (FI) developed the physics-informed spectral super-resolution framework at the core of this proposal and is first author of the paper presenting it (Haghjoo et al. 2026, arXiv:2603.18357, submitted to ApJ). The FI also built the end-to-end Roman grism simulation pipeline (extraction, contamination modeling, training-set generation from the Wang et al. 2022 IRSA products) and the Lyman- α mock-spectrum pipeline used in Task 2. The FI’s M.Sc. research (McGill University) on parameter estimation of the global 21 cm signal provides direct background in Epoch-of-Reionization physics and Bayesian inference. The FI will carry out all proposed research and lead all publications.

Prof. Bahram Mobasher (PI) is a Distinguished Professor at the University of California, Riverside, with over three decades of experience in galaxy evolution, deep multiwavelength surveys, photometric-redshift and SED-fitting methodology, and space-based survey science (HST, Spitzer, JWST, Euclid, Roman preparation). The PI supervises the FI’s PhD thesis, directs the overall research program, and mentors the FI in survey science and scientific writing.

Dr. Shoubaneh Hemmati (Collaborator, Mentor 1) is a Research Scientist at IPAC/Caltech specializing in machine-learning methods for galaxy surveys and spectroscopy; co-developer of the super-resolution framework (second author of Haghjoo et al. 2026). Dr. Hemmati advises on model architecture, validation methodology, and Roman survey spectroscopy, and co-mentors the FI (weekly meetings, plus in-person working sessions at IPAC approximately every two weeks).

3. Summary of Work Effort

Name (role)	Contribution	Effort
Aryana Haghjoo (FI)	All analysis, pipelines, publications	1.0 FTE
Bahram Mobasher (PI)	Thesis supervision, survey science guidance	no cost requested
Shoubaneh Hemmati (Mentor 1)	ML methodology co-mentoring	no cost requested

4. Facilities, Equipment, and Other Resources

Computing. The PI’s research group at UCR operates a dedicated GPU system equipped with an NVIDIA GeForce RTX 5090 (32 GB), supplemented by the University of California, Riverside High Performance Computing Center (HPCC), which provides shared GPU nodes and multi-TB project storage. Model training for this project (~ 2.5 M-parameter networks, datasets $\lesssim 10$ GB) is well within these resources; no NASA HEC allocation is required.

Data access. All observational inputs are public: JWST/NIRSpec JADES data releases (MAST) and Roman grism simulation products (Wang et al. 2022; IRSA). No letters of resource support are required for these public archives.

Simulations. Radiative-transfer post-processed hydrodynamic simulations of patchy reionization (Lyman- α transmission sightlines with matched ionization cubes) are provided through an ongoing collaboration with Prof. Anson D’Aloisio (UC Riverside); the FI already holds the simulation data and has built the training-label pipeline on them, so continued access is assured and no letter of resource support is required. Public simulation suites (THESAN, SPHINX₂₀) provide independent training sets.

5. Research Readiness Statement (Future Investigator)

I am a PhD student in Astronomy at the University of California, Riverside (enrolled since September 2024 in the M.Sc./Ph.D. track; Ph.D. program since September 2025), with expected graduation in June 2030 — fully spanning the proposed three-year award. My path to this project combines the three ingredients it requires: reionization physics, machine-learning methodology, and space-based spectroscopy.

Reionization and statistical inference. My M.Sc. in Physics at McGill University (2021–2023) was devoted to the Epoch of Reionization: I built an MCMC parameter-estimation pipeline for the global 21 cm signal using ARES simulations and EDGES data, giving me a working foundation in EoR astrophysics and Bayesian inference — the physics and statistics underlying Tasks 1 and 2.

Machine learning for spectroscopy. I am first author of “Learning to See Sharper: A Physics-Informed Artificial Intelligence Framework for Super-Resolving Galaxy Spectra” (arXiv:2603.18357, submitted to ApJ), the framework at the center of this proposal: I designed the architecture, training strategy, and validation methodology, trained it on 1,187 JADES prism–grating pairs, and released the code and checkpoints publicly. I have since ported the framework to simulated Roman grism data, building the full extraction and training pipeline (17,000+ spectral pairs to date) and the prior-dominance audit described in the S/T/M. My ML background began with undergraduate research on variable-star classification and coursework in machine learning for physics; my current graduate coursework includes galaxy formation, advanced cosmology, radiative processes, and observational astrophysics.

Observational experience. I work with JWST/NIRSpec data daily (JADES DR3/DR4); I am a Co-I of an accepted JWST Cycle 5 GO program (NIRSpec/IFU observations of extremely luminous $z \sim 9$ galaxies, unfunded role); and I have contributed to spectroscopic catalog construction from Keck data for the H2O survey. I have presented this research program at ten venues since mid-2025, including invited talks at the CfA AstroAI seminar and the Euclid US science meeting.

I am proficient in Python/PyTorch, high-performance computing (SLURM), and collaborative software development, and have prototyped every pipeline this proposal requires: the mock Lyman- α generation (Task 1), the bubble-size training-label extraction from radiative-transfer cubes (Task 2), and the Roman injection machinery (Task 3). The proposed research is the natural integration of these components into a single measurement, and I am prepared to execute it.

6. Acknowledgement of Authorship

I, Aryana Haghjoo, affirm that I am the primary author of this proposal. I defined the scientific objectives, designed the technical approach, and wrote the Science/Technical/Management section, the Open Science and Data Management Plan, and this Expertise and Resources document. The mentoring plan was prepared jointly with the PI, Prof. Bahram Mobasher, and collaborator Dr. Shoubaneh Hemmati, who also provided scientific feedback on the research plan. The proposed research builds on my first-author work developing the spectral super-resolution framework and represents my own initiative in extending it to reionization science. Portions of the text, L^AT_EX, and figure-production code were drafted and revised with the assistance of a generative AI tool (Claude, Anthropic, June–July 2026) under my direction; all scientific ideas, methods, and conclusions are my own, and I take full responsibility for the accuracy of the entire proposal.

7. Budget and Budget Narrative

Period of performance

Three years, 1 June 2027 – 31 May 2030, beginning with the Summer 2027 quarter. (The solicitation requires a start date no later than 14 July 2027; June 1, 2027 satisfies this and covers the full summer quarter.)

Year 1 (June 2027 – May 2028)

Item	Amount
FI stipend — Summer 2027 quarter: UC GSR Step 1, 50% appointment (3 mo × \$2,995.63)	\$8,987
FI stipend — Fall–Spring: UC GSR Step 3 upon advancement to candidacy, 50% (9 mo × \$3,477.96)	\$31,302
Domestic travel (AAS or one topical workshop)	\$1,500
Tuition and fee contribution	\$8,211
Total Year 1 (fellowship classification; no F&A)	\$50,000

Years 2 and 3 (each)

Item	Amount
FI stipend — 12 months at UC GSR Step 3, 50% (12 mo × \$3,477.96)	\$41,736
Domestic travel	\$1,500
Publication charges (open access, 1 paper/yr)	\$2,000
Tuition and fee contribution	\$4,764
Total per year	\$50,000
Three-year total	\$150,000

Narrative

The requested funds support the FI only; no salary, travel, or other costs are requested for the PI or collaborators, per the solicitation. The University of California, Riverside administers FINESST awardees as fellowship holders; accordingly no facilities and administrative costs are assessed and the full annual amount supports the FI directly.

The budget prioritizes the FI's stipend so that the FI is released from teaching duties for the entire award period. Stipend amounts follow the published University of California Graduate Student Researcher salary scale (UCOP Table 22, rates effective 10/1/2025: Step 1 \$5,991.25/month, Step 3 \$6,955.92/month, full-time equivalent) at a 50% appointment (20 hours per week) — the maximum employment level for the FI as an international student and the standard UC graduate appointment: Step 1 during the Summer 2027 quarter and Step 3 from Fall 2027 onward, reflecting the FI's advancement to Ph.D. candidacy. Should the UC salary scale be adjusted during the award period, the stipend will follow the prevailing scale and the tuition contribution line will absorb the difference within the annual cap.

The remaining annual amount contributes toward tuition and fees. Per the University of California 2025–26 fee schedule, graduate academic tuition and the student services fee total \$4,810 per quarter (three quarters per year; the summer quarter carries no tuition), plus campus-based fees and health insurance; nonresident supplemental tuition (\$5,034 per quarter) is assessed only until advancement to candidacy, after which it is \$0 for the duration of the award. The balance of tuition and fees not covered by this award is paid from the PI's research funds — a committed cost share that ensures the project is fully resourced within the FINESST annual cap.

Travel (\$1,500 per year) supports presentation of results at one national meeting (e.g., AAS) or one topical workshop per year. The FI's regular in-person working sessions with Dr. Hemmati at IPAC involve only local travel and require no budgeted funds. Publication charges (\$2,000 in Years 2 and 3 only) cover one open-access article per year, matching the publication schedule in the S/T/M: the Year-1 methods paper is expected to reach publication charges in Year 2, followed by the flagship measurement and Roman forecast papers.